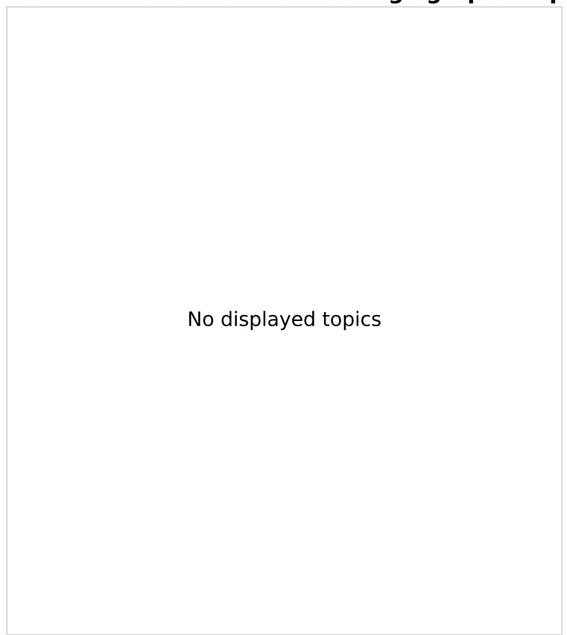
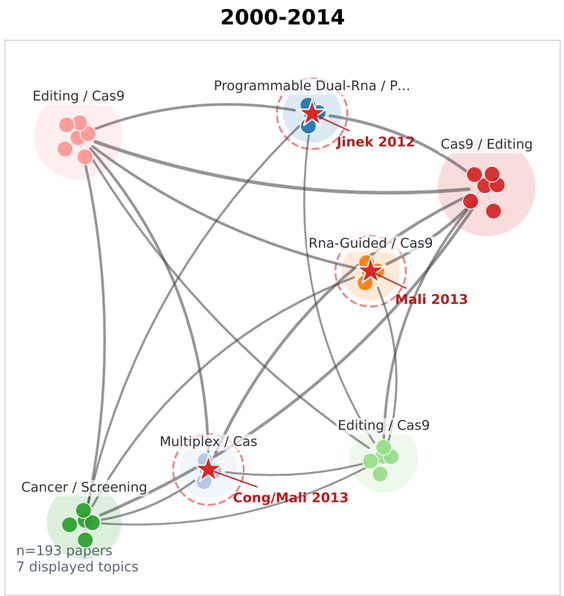
Fig. 1 | Citation knowledge-graph perturbation after a landmark innovation

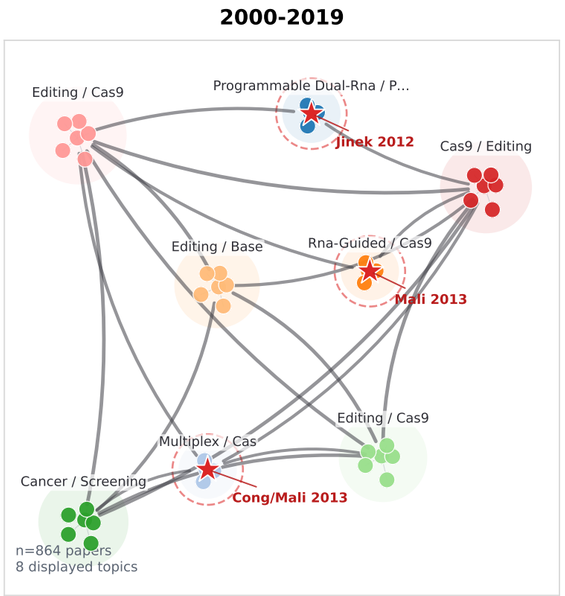
Example domain: CRISPR-Cas genome editing. Real OpenAlex data; publication-style display uses data-derived topic snapshots.

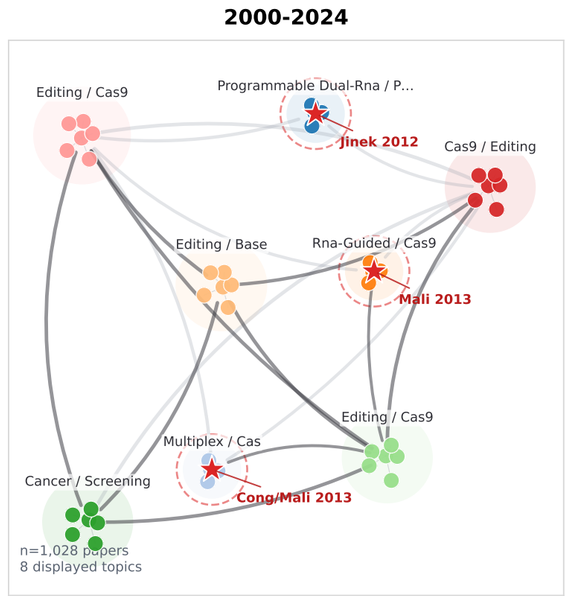
a Cumulative citation knowledge graph snapshots

CRISPR-Cas genome editing









Innovation shock: programmable Cas9 | Field reconfiguration | Compression into translational hubs

b Dominant parameter trajectories explaining the graph transitions in panel a

Bridging
B + RTD up
Cross-community paths

Novelty
Uzzi up; DeltaQ < 0
Non-typical recombination

Boundary breaking
DeltaQ < 0
Community borders disrupted

Convergence
B/RTD sustained
DeltaQ near 0